

# What Are Acids and Bases?

Definitions, conjugate pairs, strength and neutralization

Name \_\_\_\_\_ Class \_\_\_\_\_ Date \_\_\_\_\_

## A. Definitions

1. Define a Brønsted–Lowry acid and a Brønsted–Lowry base.

---

---

2. Explain the difference between a **strong** acid and a **concentrated** acid.

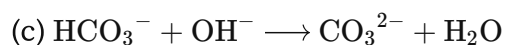
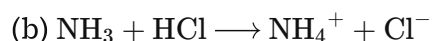
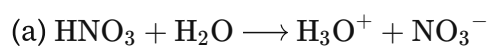
---

---

---

## B. Acids, bases and conjugate pairs

3. For each reaction, identify the acid and the base (Brønsted–Lowry):



4. Write the **conjugate base** of: (a) HF, (b)  $\text{H}_2\text{SO}_4$ , (c)  $\text{NH}_4^+$ , (d)  $\text{H}_2\text{O}$ .

5. Write the **conjugate acid** of: (a)  $\text{NH}_3$ , (b)  $\text{OH}^-$ , (c)  $\text{H}_2\text{O}$ , (d)  $\text{CO}_3^{2-}$ .

### C. Strong and weak

---

6. Classify each as a strong acid, weak acid, strong base or weak base:  $\text{HCl}$ ,  $\text{CH}_3\text{COOH}$ ,  $\text{NaOH}$ ,  $\text{NH}_3$ ,  $\text{HNO}_3$ .

### D. Neutralization

---

7. Write balanced equations for these neutralizations:

(a) hydrochloric acid with potassium hydroxide

(b) sulfuric acid,  $\text{H}_2\text{SO}_4$ , with sodium hydroxide

(c) nitric acid with calcium hydroxide,  $\text{Ca}(\text{OH})_2$

### E. Challenge

---

8. Water acts as a base in reaction 3(a) but as an acid in  $\text{NH}_3 + \text{H}_2\text{O} \rightleftharpoons \text{NH}_4^+ + \text{OH}^-$ . What is the name for a substance that can do both, and how can water act both ways?